BCS 3106

SOFTWARE ENGINEERING

SEMESTER PROJECT

GROUP B

BOBITNRB105824 - ANNETTE DALIZU

BOBITNRB152424 - BRIGHTON STANLEY

BOBITNRB336624 - MITSHELL MONYANGI

BOBITNRB323624 - MAGEL GAL DANIEL

Software Requirements Specification & System Models for Bidii Quality Builders

1. Introduction

Bidii Quality Builders is a construction company that provides house building and property improvement services. The company employs skilled workers such as bricklayers, carpenters, and plumbers, while the proprietor oversees project management and occasionally participates in construction work.

Currently, most operations including customer inquiries, cost estimation, job scheduling, material procurement, and payment processing are handled manually. This approach is inefficient, time-consuming, and prone to human error, especially as the company adopts green construction technologies and modern practices.

In today's digital era, software plays a critical role in optimizing operations. Therefore, this project proposes the development of a computerised management system to enhance efficiency, reduce operational waste, and support Lean Software Development principles.

2. Project Objective

The primary objective of this project is to design and develop a computerized management system for Bidii Quality Builders using the Lean Software Development methodology.

The specific objectives are:

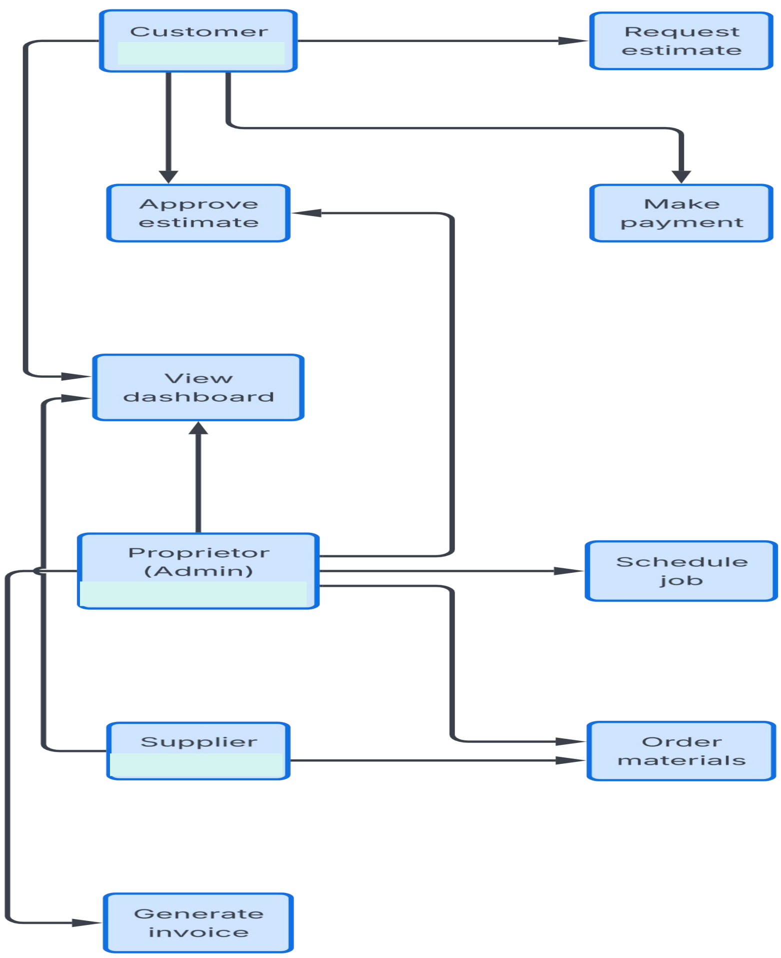
- To digitize the estimate management process: Enable customers to submit requests, allow the proprietor to record property visit details, and generate detailed estimates within 3 days of property inspection.
- To automate job scheduling: Facilitate the scheduling of jobs based on job size and existing workload, and automate the confirmation of start dates with customers.
- To streamline material procurement: Allow the proprietor to order materials from suppliers efficiently, with delivery scheduled to coincide with job start dates.
- To digitalize invoicing and payment tracking: Enable the automatic generation of invoices based on actual job costs and track customer payments within the 30-day payment period.

- To provide a centralized dashboard: Develop a web-based dashboard using Python Django that visualizes key project metrics using libraries such as Matplotlib, providing the proprietor with real-time insights into business operations.
- To apply green software engineering principles: Ensure the software is designed to be energy-efficient, minimizing computational resource usage and carbon emissions, aligning with the company's commitment to green technology.

3. System Model

3.1 Use Case Diagram

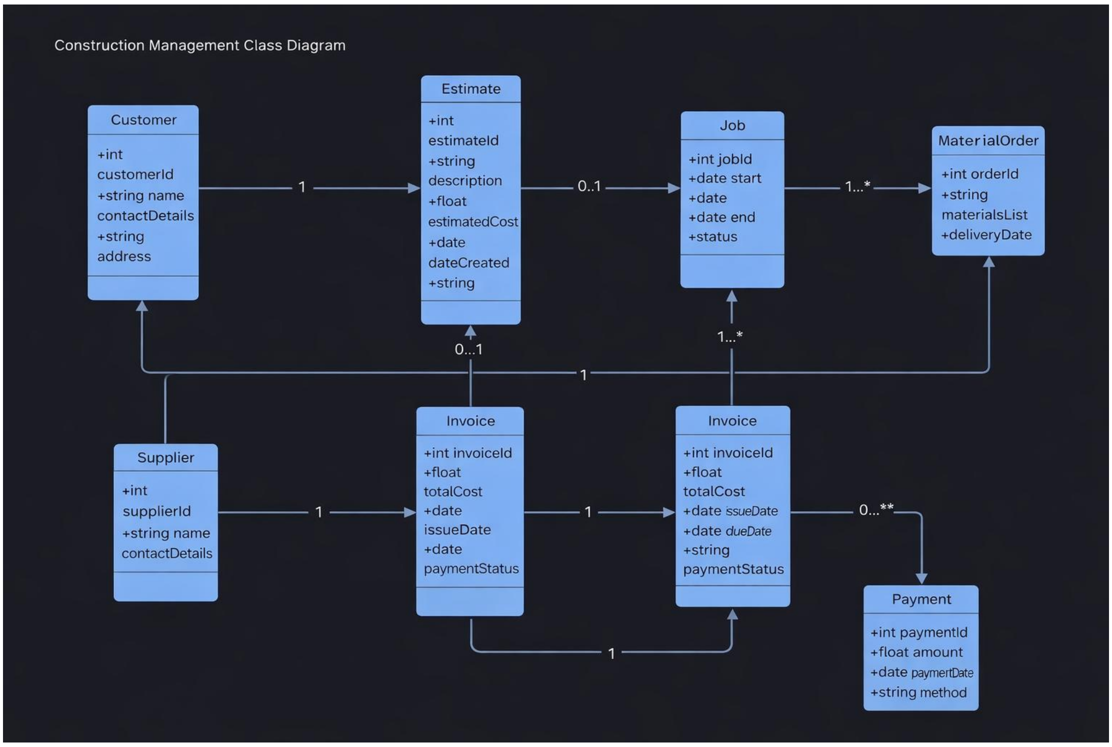
The use case diagram illustrates interactions between system users and the system.



The use case diagram above illustrates the interactions between three primary actors (Customer, Proprietor, and Supplier) and the system. The Customer can request estimates, approve estimates, and make payments. The Proprietor is the central actor who manages estimates, schedules jobs, orders materials, and generates invoices. The Supplier interacts with the system to confirm material deliveries. The diagram also implies the Worker actor who completes the job, though this is managed indirectly by the proprietor.

3.2 Class Diagram

The class diagram represents the system structure, including entities, attributes, and relationships.



The class diagram represents the static structure of the system. It includes the main entities: Customer, Proprietor, Worker, Estimate, Job, Material, Supplier, and Invoice. The relationships show that a Customer can have multiple Estimates, an Estimate leads to one Job, a Job requires multiple Materials (sourced from Suppliers), and a completed Job generates one Invoice. The Proprietor manages the entire process and oversees Workers

4. GitHub Project Link

The project is managed using GitHub for version control and collaboration.

5. Conclusion

The proposed Bidii Quality Builders Management System will significantly improve operational efficiency by automating key business processes. By adopting Lean Software Development principles and modern technologies, the system will reduce waste, enhance productivity, and support sustainable construction practices.This initial phase establishes the foundation for further development, including detailed requirements analysis, system design, and implementation.

REFERENCES

Sommerville, I. (2016). Software Engineering. 10th ed. Boston: Pearson.

Fowler, M. (2004). UML Distilled: A Brief Guide to the Standard Object Modeling Language. 3rd ed. Boston: Addison-Wesley.

Poppendieck, M., & Poppendieck, T. (2003). Lean software development: An agile toolkit. Addison-Wesley.

Kleuker, S. (2025). Grundkurs software-engineering mit UML: Der pragmatische weg zu erfolgreichen softwareprojekten (5th ed.). Springer Fachmedien Wiesbaden.